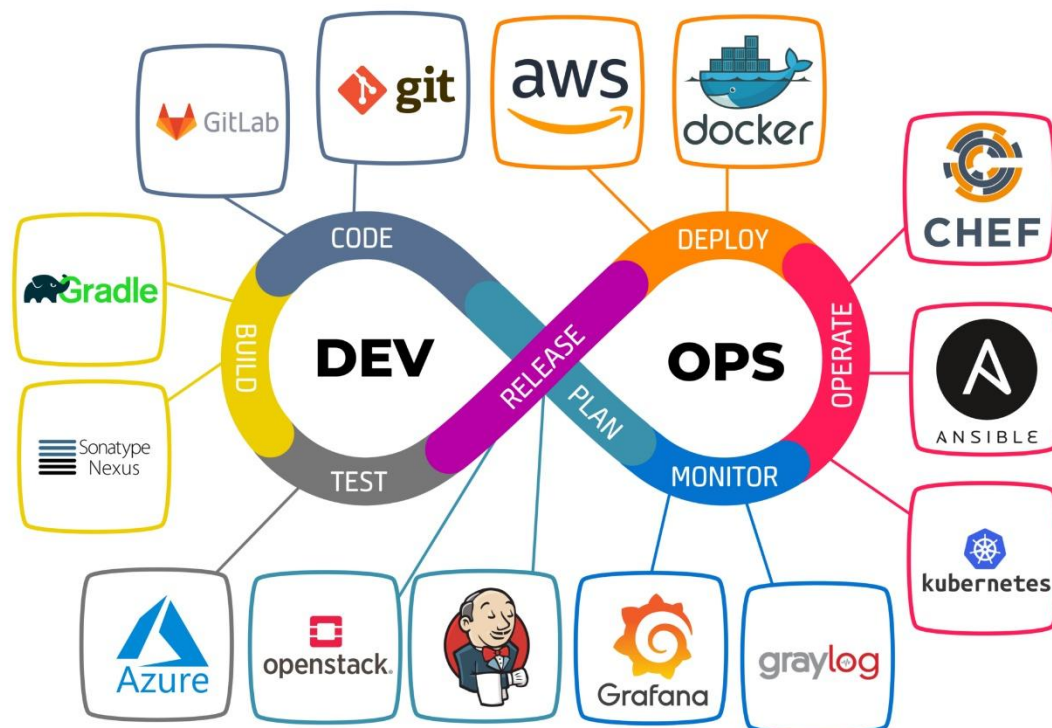


Name:-Aditi Yedge.**Roll No:-68****Div:-A****Batch:-A2****Subject:-DEVOPS**

Assignment1

Aim:- What is DevOps , key principles and practices, benefits of implementation, devops, basic git setup.



1)What is DevOps?

DevOps is a modern approach to software development that brings development and operations teams together to deliver applications faster and more reliably. It focuses on collaboration, automation, and continuous improvement across the software lifecycle.

- Encourages close collaboration between development and operations teams.
- Automates build, test, and deployment processes to reduce errors.
- Enables faster and more frequent software releases.
- Improves system reliability, monitoring, and feedback loops.

2)Key Principles

1. Collaboration

Developers and Operations work together

Breaks silos between teams

Shared responsibility

Example: Dev and Ops jointly monitor application performance.

2. Automation

Automate repetitive tasks like:

Build

Testing

Deployment

Monitoring

Tools: Jenkins, GitHub Actions, GitLab CI

Benefit: Faster and error-free deployments

3. Continuous Integration (CI)

Developers frequently merge code into shared repository

Automated tests run automatically

Goal: Detect errors early

Example: GitHub automatically runs tests after every push

4. Continuous Delivery (CD)

Code changes are automatically prepared for deployment

Continuous Delivery → manual deploy

Continuous Deployment → automatic deploy

5. Monitoring and Feedback

Continuously monitor application performance

Collect logs, metrics, alerts

Tools:

Prometheus

Grafana

ELK Stack

6. Infrastructure as Code (IaC)

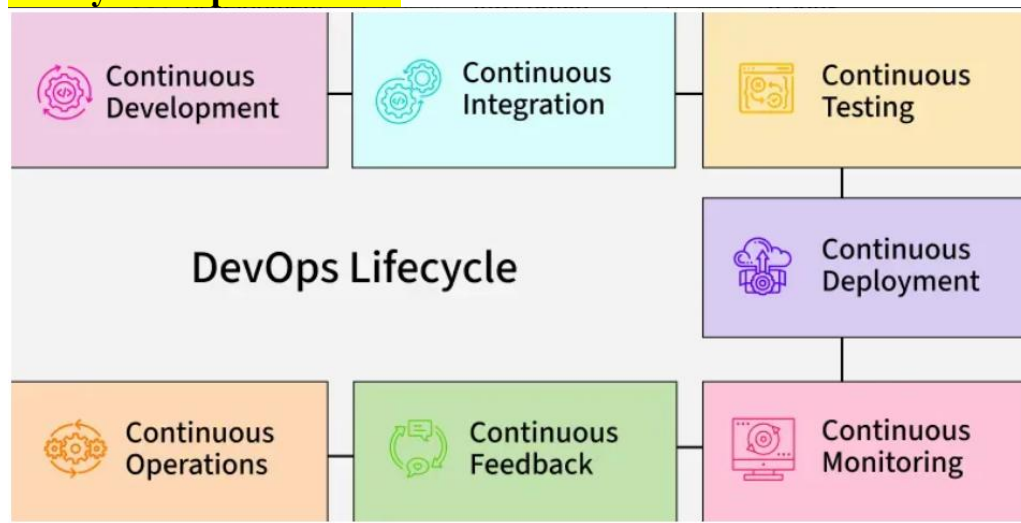
Infrastructure managed using code

Tools:

Terraform

AWS CloudFormation

3. Key DevOps Practices



1. Version Control

Track and manage code changes

Tool: Git

Example:

- GitHub
- GitLab
- Bitbucket

2. Continuous Integration (CI)

Automatically build and test code

Tools:

- Jenkins
- GitHub Actions

3. Continuous Deployment (CD)

Automatically deploy applications

Tools:

- Jenkins
- ArgoCD

4. Containerization

Package application with dependencies

Tool: Docker

Benefit:

- Runs same everywhere

5. Configuration Management

Manage servers automatically

Tools:

- Ansible
- Puppet
- Chef

6. Monitoring and Logging

Track system health

Tools:

- Prometheus
- Grafana
- ELK Stack

4) Benefits of implementing devops

1. Faster Software Delivery

Explanation

DevOps enables teams to release software more frequently and quickly by automating build, test, and deployment processes.

Traditional model:

- Releases happen once a month or once in several months.

DevOps model:

- Releases can happen daily or multiple times per day.

How DevOps achieves this:

- Continuous Integration (CI)
- Continuous Deployment (CD)
- Automated pipelines

Example:

Amazon deploys code thousands of times per day using DevOps.

Benefit:

- Faster delivery of new features
 - Faster bug fixes
-

2. Improved Collaboration

Explanation

In traditional systems:

- Developers write code
- Operations deploy and maintain systems
- Communication gaps exist

DevOps removes this barrier.

DevOps approach:

- Developers and Operations work together
- Shared responsibility
- Better teamwork

Result:

- Faster problem solving
 - Better communication
-

3. Higher Quality Software

Explanation

DevOps uses automated testing tools to detect bugs early.

Types of automated testing:

- Unit testing
- Integration testing
- System testing

Tools used:

- Selenium
- JUnit
- TestNG

Benefit:

- Fewer bugs in production
 - More reliable software
-

4. Faster Problem Resolution**Explanation**

DevOps includes continuous monitoring of systems.

Monitoring tools detect issues immediately.

Tools:

- Prometheus
- Grafana
- ELK Stack

Example:

If server crashes, alert is generated immediately.

Benefit:

- Faster detection
 - Faster fixing
-

5. Increased Reliability**Explanation**

Automation reduces human errors.

Manual deployments can cause mistakes.

DevOps uses:

- Automated deployment scripts
- Automated testing

Result:

- Stable systems
 - Consistent deployments
-

6. Scalability**Explanation**

DevOps supports Infrastructure as Code (IaC).

Infrastructure can be scaled automatically.

Tools:

- Terraform
- AWS CloudFormation

Example:

Automatically increase servers when traffic increases.

Benefit:

- Handle large number of users easily
-

7. Better Customer Satisfaction**Explanation**

DevOps ensures:

- Faster updates
- Fewer bugs
- Reliable systems

Customers receive improvements faster.

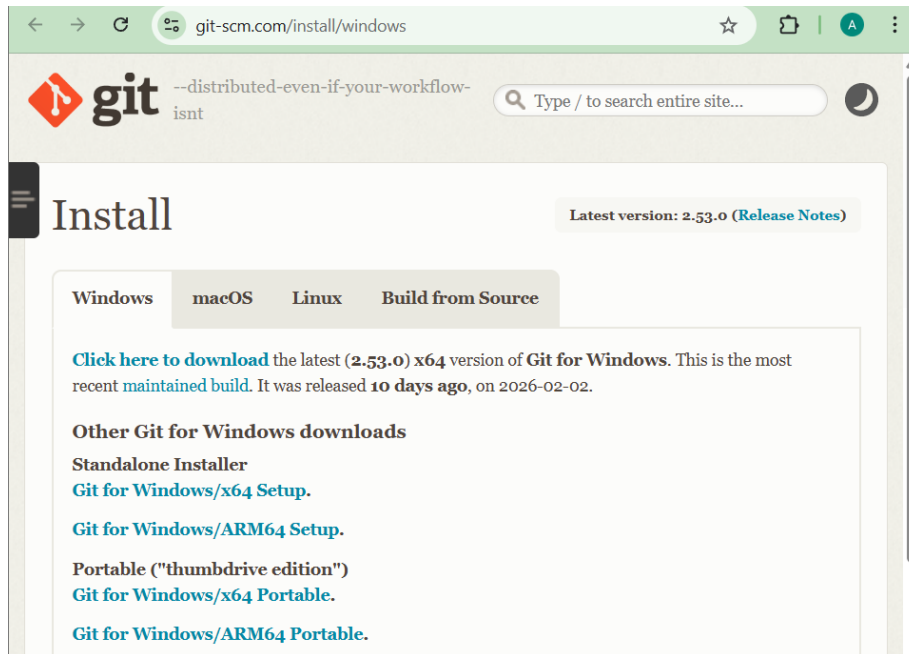
Result:

- Better user experience
- Increased customer trust

5) basic git setup

Step 1: Download the Installer.

Go to the official Git website: <https://git-scm.com/downloads/win>



The download will start automatically for the latest version of Git for Windows. After the download is complete, run the .exe file.

Step 2.: Complete the installation.

1)To check the version of git

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements!

PS D:\MCA_SEM_II\DEVOPS> git --version
git version 2.50.1.windows.1
```

2) Configure Git User Information

- `git config --global user.name "Aditi"`
- `git config --global user.email "aditiyedge10@gmail.com"`

```
MINGW64:/c/Users/Admin

Admin@DESKTOP-JM2K4MP MINGW64 ~
$ git --version
git version 2.52.0.windows.1

Admin@DESKTOP-JM2K4MP MINGW64 ~
$ git config --global user.name "Aditi"

Admin@DESKTOP-JM2K4MP MINGW64 ~
$ git config --global user.email "aditiyedge10@gmail.com"
```

3) Navigate to Project folder and initialize a Git repository.

```
MINGW64:/d/MCA_SEM_II/DEVOPS

Adii@DESKTOP-C60SFL6 MINGW64 ~
$ cd /d/MCA_SEM_II/DEVOPS

Adii@DESKTOP-C60SFL6 MINGW64 /d/MCA_SEM_II/DEVOPS
$ git init
Initialized empty Git repository in D:/MCA_SEM_II/DEVOPS/.git/

Adii@DESKTOP-C60SFL6 MINGW64 /d/MCA_SEM_II/DEVOPS (master)
$ git add .

Adii@DESKTOP-C60SFL6 MINGW64 /d/MCA_SEM_II/DEVOPS (master)
$ git add Aditi.txt

Adii@DESKTOP-C60SFL6 MINGW64 /d/MCA_SEM_II/DEVOPS (master)
$ cat Aditi.txt
Hello World!

Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ vim pract1.txt

Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ ..
```

4) Display the contents of the file

- cat pract1.txt

```
Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ cat pract1.txt
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```

5) Check Repository Status

- git status

```
Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ git status
warning: could not open directory 'Application Data/': Permission denied
warning: could not open directory 'Cookies/': Permission denied
warning: could not open directory 'Local Settings/': Permission denied
warning: could not open directory 'My Documents/': Permission denied
warning: could not open directory 'NetHood/': Permission denied
warning: could not open directory 'PrintHood/': Permission denied
warning: could not open directory 'Recent/': Permission denied
warning: could not open directory 'SendTo/': Permission denied
warning: could not open directory 'Start Menu/': Permission denied
warning: could not open directory 'Templates/': Permission denied
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   pract1.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .VirtualBox/
    .anaconda/
    .conda/
    .continuum/
    .eclipse/
    .gitconfig
    .ipynb_checkpoints/
    .ipython/
    .kettle/
    .lemminx/
    .m2/
    .p2/
    .pentaho/
    .swt/
    .tooling/
    .viminfo
    .vscode/
    A15/
    AppData/
    Contacts/
    Desktop/
    Documents/
    Downloads/
    Favorites/
    Links/
    Music/
    NTUSER.DAT
    NTUSER.DAT{1c2b59c5-c5f5-11eb-bacb-000d3a96488e}.TxR.0.regtrans-ms
    NTUSER.DAT{1c2b59c5-c5f5-11eb-bacb-000d3a96488e}.TxR.1.regtrans-ms
```

6) Commit Changes to the Repository

- git commit -m "fistCommit"

```
Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ git commit -m fistCommit
[master (root-commit) 426c896] fistCommit
1 file changed, 1 insertion(+)
create mode 100644 pract1.txt
```


7) Manage Branches (Create, List, and Delete)

- `git branch Div_A68`
- `git branch`
- `git branch -d Div_A68`

```
Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ git branch Div_A68

Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ git branch
Div_A68
* master

Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ git branch -d Div_A68
Deleted branch Div_A68 (was 426c896).

Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ git branch
* master
```

8) Checkout Commands (Switching and Restoring)

- `git checkout master`
- `git checkout pract1.txt`
- `git checkout -b Div_A68`

```
Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ git checkout master
Already on 'master'

Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ git checkout pract1.txt
Updated 0 paths from the index

Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ git checkout -b Div_A68
Switched to a new branch 'Div_A68'
```

9) View Commit History

- `git log`

```
Admin@DESKTOP-JM2K4MP MINGW64 ~ (Div_A68)
$ git log
commit 426c896ee2e66f300767e4dafa5c7740f476e5a2 (HEAD -> Div_A68, master)
Author: Aditi <aditiyedge10@gmail.com>
Date: Thu Feb 5 21:07:57 2026 -0800

    fistCommit
```

10) Switching Branches

- `git switch master`

- `git switch -c divA`

```
Admin@DESKTOP-JM2K4MP MINGW64 ~ (Div_A68)
$ git switch master
Switched to branch 'master'

Admin@DESKTOP-JM2K4MP MINGW64 ~ (master)
$ git switch -c divA
Switched to a new branch 'divA'
```

11) Merging Branches

- `git merge master`

```
Admin@DESKTOP-JM2K4MP MINGW64 ~ (divA)
$ git merge master
Already up to date.
```

12) View Changes and Configure Credentials

- `git config --global credential.helper cache`
- `git show master`

```
Admin@DESKTOP-JM2K4MP MINGW64 ~ (divA)
$ git config --global credential.helper cache
```

```
Admin@DESKTOP-JM2K4MP MINGW64 ~ (divA)
$ git show master
commit 426c896ee2e66f300767e4dafa5c7740f476e5a2 (HEAD -> divA, master, Div_A68)
Author: Aditi <aditiyedge10@gmail.com>
Date: Thu Feb 5 21:07:57 2026 -0800

    fistCommit

diff --git a/pract1.txt b/pract1.txt
new file mode 100644
index 0000000..9d6f716
--- /dev/null
+++ b/pract1.txt
@@ -0,0 +1 @@
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DevOps Lab
Admin@DESKTOP-JM2K4MP MINGW64 ~ (divA)
$ git show divA
```